2201NSC Linear Algebra and Applications

Week 7

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Vector Spaces

Subspaces, bases, dimension and matrix spaces

Vectors need not be columns of numbers

A vector space is a collection of objects that can be added and multiplied by scalars while obeying the familiar algebraic rules.

Coordinate vectors	Matrices	Functions
$\mathbb{R}^n$	$\mathbb{R}^{m \times n}$	Polynomials, continuous functions and solutions of homogeneous linear ODEs.

Once the rules hold, every object in the space is called a **vector**.

Basis functions turn a function into a coordinate vector

Choose basis functions

$$\mathcal{B} = \{f_1, f_2, \ldots, f_n\}.$$

If

$$f = c_1 f_1 + c_2 f_2 + \cdots + c_n f_n,$$

then f is represented by its coefficients:

$$\boxed{[f]_{\mathcal{B}} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}.}$$

For $p(x) = 2 - 3x + 5x^2$ and $\mathcal{B} = \{1, x, x^2\}$,

$$p = 2(1) - 3(x) + 5(x^2) \qquad \Longleftrightarrow \qquad \boxed{[p]_{\mathcal{B}} = \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix}.}$$

Vector spaces are closed under both operations

For all $\mathbf{x}, \mathbf{y} \in V$ and every scalar $\alpha \in \mathbb{R}$, closure requires

$$\boxed{\mathbf{x} + \mathbf{y} \in V} \qquad \text{and} \qquad \boxed{\alpha \mathbf{x} \in V.}$$

Why closure matters

Every linear combination $\alpha \mathbf{x} + \beta \mathbf{y}$ must remain in the space.

Closure is necessary, but not sufficient

A vector space must obey eight additional algebraic rules, shown on the next slide.

The remaining rules preserve familiar algebra

For $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$ and $\alpha, \beta \in \mathbb{R}$:

Vector addition

$$\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$$
$$(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$$
$$\mathbf{x} + \mathbf{0} = \mathbf{x}$$
$$\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$$

Scalar multiplication

$$\alpha(\mathbf{x} + \mathbf{y}) = \alpha\mathbf{x} + \alpha\mathbf{y}$$
$$(\alpha + \beta)\mathbf{x} = \alpha\mathbf{x} + \beta\mathbf{x}$$
$$(\alpha\beta)\mathbf{x} = \alpha(\beta\mathbf{x})$$
$$1\mathbf{x} = \mathbf{x}$$

A subspace is a vector space inside a vector space

Let S be a nonempty subset of a known vector space V. Then S is a **subspace** of V when

$$\boxed{\mathbf{x}, \mathbf{y} \in S \implies \mathbf{x} + \mathbf{y} \in S}$$

and

$$\boxed{\mathbf{x} \in S,\ \alpha \in \mathbb{R} \implies \alpha\mathbf{x} \in S.}$$

The other vector-space rules are inherited from V, so only closure needs to be checked.

The zero vector gives a fast subspace test

Let

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \mathbb{R}^3, \qquad \mathbf{0} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Every subspace of $\mathbb{R}^3$ must contain $\mathbf{0}$ because $0\mathbf{x} = \mathbf{0}$.

S_0: homogeneous condition

$$S_0 = \{\mathbf{x} \in \mathbb{R}^3 : x_1 + x_3 = 0\}$$

For $\mathbf{x} = \mathbf{0}$, $x_1 + x_3 = 0 + 0 = 0$. Thus $\mathbf{0} \in S_0$ and closure may hold.

S_1: nonhomogeneous condition

$$S_1 = \{\mathbf{x} \in \mathbb{R}^3 : x_1 + x_3 = 1\}$$

For $\mathbf{x} = \mathbf{0}$, $x_1 + x_3 = 0 \neq 1$. Thus $\mathbf{0} \notin S_1$, so S_1 cannot be a subspace.

A null space contains all homogeneous solutions

For $A \in \mathbb{R}^{m \times n}$,

$$\boxed{\mathcal{N}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}.}$$

The null space is a subspace of $\mathbb{R}^n$ because

$$A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y} = \mathbf{0}$$

and

$$A(\alpha \mathbf{x}) = \alpha A\mathbf{x} = \mathbf{0}$$

whenever $\mathbf{x}, \mathbf{y} \in \mathcal{N}(A)$.

The zero solution guarantees that $\mathcal{N}(A)$ is nonempty.

The span contains every reachable linear combination

For vectors $\mathbf{v}_1, \ldots, \mathbf{v}_k$,

$$\boxed{\operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\} = \{c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k : c_i \in \mathbb{R}\}\,.}$$

Examples include

$$\mathbb{R}^3 = \operatorname{span}\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$$

and

$$\mathcal{P}_4 = \operatorname{span}\{1, x, x^2, x^3\},$$

where $\mathcal{P}_4$ contains polynomials of degree at most 3.

Free variables generate a basis for the null space

Let

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix} \quad \longrightarrow \quad \operatorname{rref}(A) = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

The pivot variables are x_1 and x_2, so x_3 is free. Solving $A\mathbf{x} = \mathbf{0}$ gives

$$x_2 = 0, \qquad x_1 = -x_3.$$

Set $x_3 = t$:

$$\mathbf{x} = t \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}.$$

Therefore

$$\boxed{\mathcal{N}(A) = \operatorname{span}\left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}.}$$

A basis spans without redundancy

The vectors $\mathbf{v}_1, \ldots, \mathbf{v}_k$ form a **basis** for V when

1. they span V;

2. they are linearly independent.

Every $\mathbf{x} \in V$ then has a unique representation

$$\boxed{\mathbf{x} = c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k.}$$

Two roles

Spanning guarantees existence of coordinates; linear independence guarantees their uniqueness.

Bases describe matrices and polynomials with coordinates

Matrices in $\mathbb{R}^{2\times 2}$

The standard basis is

$$E_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad E_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

$$E_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad E_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Polynomials in $\mathcal{P}_3$

$$\{1, x, x^2\}$$

is the standard basis. The polynomial

$$2 - 3x + 5x^2$$

has coordinate vector $(2, -3, 5)^{\mathsf{T}}$.

Dimension counts the vectors in a basis

Every basis for a finite-dimensional space contains the same number of vectors. This number is the **dimension**.

$\mathbb{R}^n$

$$\dim(\mathbb{R}^n) = n$$

$\mathbb{R}^{m \times n}$

$$\dim(\mathbb{R}^{m \times n}) = mn$$

$\mathcal{P}_k$

Polynomials of degree less than k have dimension k.

In an n-dimensional space, any n linearly independent vectors form a basis.

Column and row spaces come from a matrix's vectors

For $A \in \mathbb{R}^{m \times n}$,

$$\boxed{\mathrm{Col}(A) = \text{span of the columns of } A \subseteq \mathbb{R}^m}$$

and

$$\boxed{\mathrm{Row}(A) = \text{span of the rows of } A \subseteq \mathbb{R}^n.}$$

If $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$, then

$$A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n.$$

Thus the column space is precisely the set of all possible outputs $A\mathbf{x}$.

Consistency is membership in the column space

The system

$$A\mathbf{x} = \mathbf{b}$$

is consistent exactly when $\mathbf{b}$ can be written as a linear combination of the columns of A:

$$\boxed{A\mathbf{x} = \mathbf{b} \text{ has a solution} \iff \mathbf{b} \in \mathrm{Col}(A).}$$

For $A \in \mathbb{R}^{m \times n}$, the system is consistent for every $\mathbf{b} \in \mathbb{R}^m$ exactly when

$$\mathrm{Col}(A) = \mathbb{R}^m.$$

Use original columns but reduced rows

After row reducing A:

Basis for $\mathrm{Col}(A)$

Identify the pivot columns in the reduced matrix, then select the corresponding columns from the **original matrix** A.

Basis for $\mathrm{Row}(A)$

Use the nonzero rows of the **reduced matrix**.

Do not interchange these rules

Row operations preserve row space, but generally change the column space.

Rank and nullity count different freedoms

For $A \in \mathbb{R}^{m \times n}$,

Rank

$$\operatorname{rank}(A) = \dim \operatorname{Col}(A) = \dim \operatorname{Row}(A)$$

It equals the number of pivots.

Nullity

$$\operatorname{nullity}(A) = \dim \mathcal{N}(A)$$

It equals the number of free variables.

Both quantities can be read from a row-reduced matrix.

Rank plus nullity equals the number of columns

If A has n columns, every variable is either a pivot variable or a free variable. Therefore,

$$\boxed{\text{rank}(A) + \text{nullity}(A) = n.}$$

For example, if a 3×5 matrix has rank 3, then

$$\text{nullity}(A) = 5 - 3 = 2.$$

Interpretation

Rank counts independent output directions; nullity counts independent input directions that produce zero.

Column space and nullity classify the solutions

For $A\mathbf{x} = \mathbf{b}$:

No solution

$$\mathbf{b} \notin \mathrm{Col}(A)$$

Unique solution

$$\mathbf{b} \in \mathrm{Col}(A), \quad \mathcal{N}(A) = \{\mathbf{0}\}$$

Infinitely many

$$\mathbf{b} \in \mathrm{Col}(A), \quad \mathrm{nullity}(A) > 0$$

If $\mathbf{x}_p$ is one solution, every solution has the form

$$\boxed{\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h, \qquad \mathbf{x}_h \in \mathcal{N}(A).}$$

Null spaces can detect errors in transmitted data

Using arithmetic modulo 2, coding theory chooses valid binary messages from the null space of a check matrix H:

$$\boxed{H\mathbf{v} = \mathbf{0}.}$$

If one bit is changed, the received vector becomes

$$\mathbf{v} + \mathbf{e}_i,$$

so

$$H(\mathbf{v} + \mathbf{e}_i) = H\mathbf{e}_i,$$

which is the ith column of H.

Distinct nonzero columns allow the position of a single-bit error to be identified and corrected.

Short Revision Quiz

Subspaces, bases, dimensions and matrix spaces

Question 1 — Closure under linear combinations

If V is a vector space and $\mathbf{x}, \mathbf{y} \in V$, which expression must also belong to V?

A. $\alpha\mathbf{x} + \beta\mathbf{y}$ for all $\alpha, \beta \in \mathbb{R}$

B. $\mathbf{x}^\mathsf{T}\mathbf{y}$

C. $\mathbf{x}^{-1} + \mathbf{y}^{-1}$

Question 2 — Polynomial vector spaces

Which set is a vector space under the usual polynomial operations?

A. All polynomials of degree exactly 2

B. All polynomials of degree at most 2

C. All polynomials with constant term 1

Question 3 — Subspace test

Why is

$$S = \{\mathbf{x} \in \mathbb{R}^3 : x_2 - x_3 = 2\}$$

not a subspace of $\mathbb{R}^3$?

A. It does not contain the zero vector

B. It contains too many vectors

C. It contains vectors with negative entries

Question 4 — Subspace requirements

Let $S \subseteq V$, where V is a vector space. What is required for S to also be a vector space under the same operations?

A. S must contain exactly $\dim(V)$ vectors

B. S must be nonempty and closed under addition and scalar multiplication

C. Every vector in V must also belong to S

Question 5 — Null space

If A is an $m \times n$ matrix, then $\mathcal{N}(A)$ is a subspace of:

A. $\mathbb{R}^m$

B. $\mathbb{R}^n$

C. $\mathbb{R}^{m \times n}$

Question 6 — Span

What is $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$?

A. All linear combinations $c_1\mathbf{v}_1 + c_2\mathbf{v}_2$

B. Only $\mathbf{v}_1 + \mathbf{v}_2$

C. Only the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$

Question 7 — Basis

A basis for V must be:

A. Orthogonal and normalised

B. Linearly independent and spanning

C. The standard coordinate vectors

Question 8 — Dimension

What is the dimension of the space of polynomials of degree at most 3?

A. 3

B. 4

C. Infinite

Question 9 — Coordinates

Relative to the ordered basis $\{1, x, x^2\}$, what is the coordinate vector of $2 + x + x^2$?

A. $(1, 1, 2)^\mathsf{T}$

B. $(2, 1, 1)^\mathsf{T}$

C. $(2, 0, 1)^\mathsf{T}$

Question 10 — Column-space basis

After row reduction identifies the pivot columns, where should the basis vectors for $\mathrm{Col}(A)$ be taken from?

A. The corresponding columns of the original matrix

B. The corresponding columns of the reduced matrix

C. The nonzero rows of the original matrix

Question 11 — Row-space basis

After row reducing a matrix, where should the basis vectors for Row(A) be taken from?

A. The pivot rows of the original matrix

B. The nonzero rows of the reduced matrix

C. The pivot columns of the reduced matrix

Question 12 — Find a column-space basis

Given

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 1 & 2 & 2 \end{pmatrix}, \qquad \mathrm{rref}(A) = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},$$

which is a basis for $\mathrm{Col}(A)$?

A. $\left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} \right\}$

B. $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$

C. $\left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \\ 2 \end{pmatrix} \right\}$

Question 13 — Find a row-space basis

Given

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 1 & 2 & 2 \end{pmatrix}, \qquad \mathrm{rref}(A) = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},$$

which is a basis for $\mathrm{Row}(A)$?

A. $\{(1,2,1),(2,4,2)\}$

B. $\{(1,2,0),(0,0,1)\}$

C. $\{(1,0,0),(0,1,0)\}$

Question 14 — Rank–nullity

A matrix has five columns and rank 3. What is its nullity?

A. 2

B. 3

C. 5

Quiz answers 1–5

1 — A

Closure under addition and scalar multiplication keeps every linear combination in V.

2 — B

Polynomials of degree at most 2 include the zero polynomial and are closed.

3 — A

$\mathbf{0}$ does not satisfy $x_2 - x_3 = 2$.

4 — B

S must be nonempty and closed under addition and scalar multiplication.

5 — B

$A\mathbf{x} = \mathbf{0}$ has n unknowns, so $\mathcal{N}(A) \subseteq \mathbb{R}^n$.

Quiz answers 6–10

6 — A

The span is the set of every linear combination of the listed vectors.

7 — B

A basis is a linearly independent spanning set.

8 — B

$\{1, x, x^2, x^3\}$ is a basis, so the dimension is 4.

9 — B

$2 + x + x^2$ has coefficients $(2, 1, 1)$ in the stated order.

10 — A

Use pivot locations from the reduction but take columns from the original matrix.

Quiz answers 11–14

11 — B

The nonzero rows of the reduced matrix form a basis for $\mathrm{Row}(A)$.

12 — A

The pivot columns are 1 and 3, so take columns 1 and 3 from the original matrix.

13 — B

The nonzero rows $(1, 2, 0)$ and $(0, 0, 1)$ of the RREF form a row-space basis.

14 — A

Rank–nullity gives $3 + \mathrm{nullity}(A) = 5$.

Final takeaway

Vector spaces organise linear combinations; bases provide coordinates; and rank and nullity describe the behaviour of linear systems.